

ARSALAN AHMED

🌐 California, United States 📞 (510) 766-4922 ✉ arsalandoha@gmail.com 🔗 [linkedin.com/in/arsalanprofile](https://www.linkedin.com/in/arsalanprofile)

Experience

Test Engineering Intern

Jul 2026 – Present

Renesas Electronics

San Jose, CA

- Supported development and debug of automated test programs for analog, mixed-signal, and power management devices on ATE platforms (Advantest V93000 / COHU Diamondx).
- Wrote and modified test code in C++ and Python to support characterization and production test flows under guidance of senior test engineers.
- Conducted bench characterization using oscilloscopes, source measure units, and power supplies, then processed test data (STDF, CSV, Excel) to identify yield anomalies and correlate results with ATE measurements.

Microfabrication Research Intern

Jun 2025 – Sep 2025

NorthWest AI Hub

Davis, CA

- Operated in a cleanroom environment to fabricate semiconductor devices using etching, alignment, exposure, and deposition techniques, validating process parameters using probing systems.
- Characterized electrical devices through measurements and scanning electron microscopy, analyzing device structure and performance against theoretical models.
- Planned and executed photovoltaic device testing experiments on solar cell materials and doping profiles, assessing performance against design requirements and documenting results in structured reports.

Robotics Production Intern

Jun 2024 – Jun 2025

Barobo Inc.

Davis, CA

- Designed and contributed to optimize PCB layouts to improving circuit efficiency and compatibility.
- Assembled and soldered **1000+** circuit boards and robots; developed and executed functional verification test procedures prior to shipment.
- Streamlined testing, debugging, and quality inspection processes for electronic kits, reducing defect rates through systematic fault isolation and agile documentation.

Education

University of California, Davis

Jun 2025

Bachelor of Science, Electrical Engineering; Relevant Coursework: Integrated Circuit Fabrication, Device Physics *Davis, CA*

Projects

Autonomous Vehicle Design

OpenMV, PWM, Custom PCB

- Designed and integrated a full autonomous RC vehicle system with camera-based lane detection, translating performance requirements into an end-to-end model.
- Designed and tested a custom motor control PCB using H-Bridge and MOSFET drivers for DC motor control; used oscilloscopes to probe and validate PCB signals.
- Implemented filtered position/angle tracking and adaptive speed control, achieving consistent lap completion with a best time of **48 seconds**.

Deep Learning Vehicle Classifier

PyTorch, ResNet18

- Developed a deep learning vehicle classification system, implementing a custom PyTorch Dataset class, data augmentation techniques, and transfer learning with ResNet18.
- Optimized model performance through hyperparameter tuning (learning rate, batch size) and advanced visualization techniques, including confusion matrices and confidence-based prediction analysis.

Multi-Object Tracking System

YOLOv3

- Integrated YOLOv3 for object detection and centroid extraction, enabling real-time identification and tracking of multiple objects within video frames.
- Developed a centroid-based tracking system with unique ID assignment, object registration, and trajectory visualization for multi-object tracking.
- Achieved a **30% reduction** in tracking errors by refining object association algorithms.

Technical Skills

Languages: C++, Python, HTML, MATLAB, LaTeX, Verilog, RISC-V, SQL, Linux Command line

Developer Tools: Visual Studio, Jupyter Notebook, Eclipse IDE, Silvaco, Intel Quartus, Microsoft Office

Technologies&Frameworks: React JS, Next.js, Vercel, PyTorch, ResNet18, YOLOv3, Altium, Revit